

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising Waters – AI-Powered Flood Prediction and Early Warning System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Interface	Allows users to enter Flood Risk-related parameters	Frontend/UI	Done	High
2	Data Validation	Validates user input values	Backend	Done	Medium
3	Data Preprocessing	Processes input data before prediction	ML Module	Done	High
4	Flood Prediction	Predicts Human Development Index value	Prediction Engine	Done	High
5	Development Category Classification	Classifies predicted Flood Risk into categories	ML Module	Done	medium
6	Result Display	Displays prediction results to users	Frontend/UI	Done	Medium
7	Flask Web Application	Integrates frontend with ML model	Application Layer	Done	High
8	Model Loading & Execution	Loads trained model and performs inference	Backend	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8

Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input Form, Prediction Result Page
2	Backend / Logic	4	Input Validation, Data Preprocessing, Flood Prediction, Classification
3	Database / Storage	2	Dataset Storage, Trained Model File (.pkl)
4	API / Integration	1	Flask Application Integration
5	Security / Authentication	0	Not Implemented